

Project Title:

SmartGreen Dock: A Data-Driven Urban Biodiversity and Community Engagement Pilot

Theme:

Urban Greening & Biodiversity

Summary (Executive Overview)

SmartGreen Dock is a six-month pilot that aims to strengthen urban biodiversity, environmental awareness, and community participation across Dublin's Docklands. The project will install smart, sensor-enabled green spaces that gather continuous environmental data to guide local action. Using a network of IoT devices, the system will track ecological indicators such as soil moisture, air quality, temperature, and pollution. These insights will be visualized through AI-assisted tools, turning complex data into clear information for residents, schools, and city planners.

At its core, SmartGreen Dock is about connecting people with the living systems that surround them. A smart app—available on web and mobile—will let community members view live environmental conditions, record biodiversity observations, and take part in local greening and monitoring activities. Through this participatory model, the project will test how technology and shared knowledge can foster collaboration, civic pride, and stewardship of public spaces.

By blending digital innovation with civic engagement, SmartGreen Dock will demonstrate a model for climate-resilient city planning. The pilot will explore how real-time data can guide greener infrastructure, strengthen pollinator habitats, reduce heat-island effects, and create thriving spaces where technology and ecology support each other. The long-term goal is to deliver measurable environmental improvements, enhance well-being, and give Dublin City Council a scalable framework for sustainable, data-driven policy.

Objectives

1. Monitor key environmental and biodiversity indicators—temperature, humidity, air quality, and pollution—using an affordable IoT sensor network.
 2. Build a living digital map of biodiversity data accessible to local residents, schools, and city planners.
 3. Encourage citizen participation through a simple web/mobile platform for logging sightings, caring for plants, and sharing observations.
 4. Translate environmental data into insights that inform Dublin City Council's climate and biodiversity actions.
 5. Measure the social impact through engagement metrics, surveys, and participant feedback.
 6. Test the use of environmental drones for monitoring and managing sustainable development across the area.
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Innovation & Impact

Innovation: SmartGreen Dock brings together IoT sensors, citizen science, and interactive data visualisation within one connected platform. It demonstrates how a mix of smart infrastructure and human participation can create an intelligent, community-driven sustainability model.

Impact:

- Broader biodiversity awareness and hands-on learning for residents and schools.
- Better local environmental conditions, from improved soil moisture to cooler microclimates.
- A functioning prototype for a scalable citywide greening strategy.
- Stronger data evidence to support local and national policy decisions.

Success Metrics:

- Number of active participants using the app or taking part in workshops.
 - Quantity and quality of biodiversity data gathered.
 - Observable improvements in environmental indicators such as pollinator counts or vegetation health.
 - Operation of smart green micro-sites within the Docklands.
 - Launch of a public “SmartGreen Dock Dashboard.”
 - Engagement of at least 150 residents or students.
 - Completion of an impact report and a roadmap for citywide adoption.
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Scalability & Sustainability

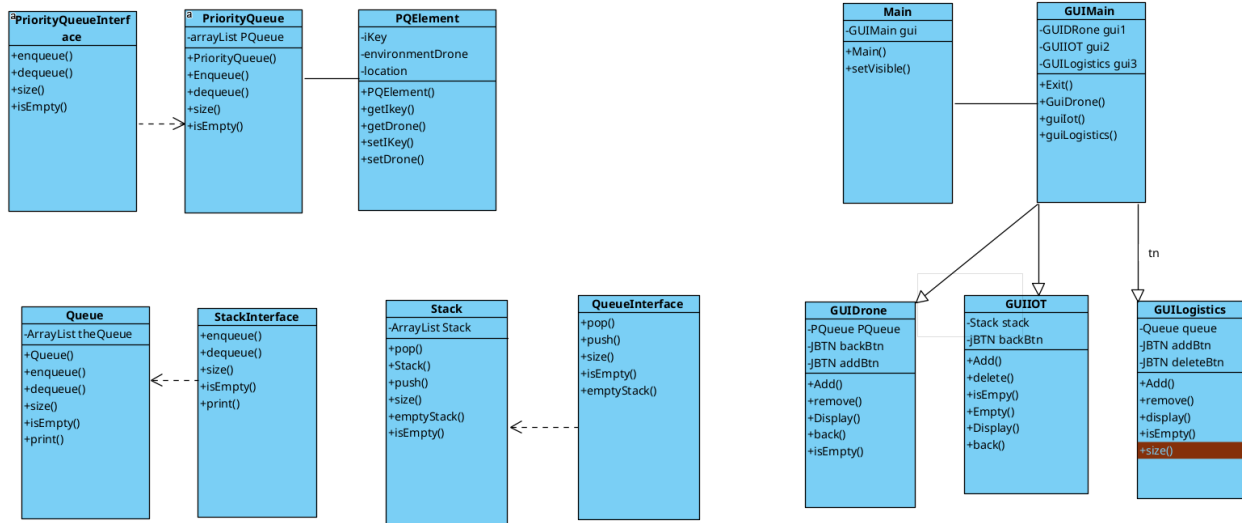
After the pilot, the framework can expand through:

- Gradual extension of the IoT network into additional districts.
 - Insights feeding directly into Dublin’s Climate Action and Biodiversity Strategies.
 - Development of a shared “drone hub” to coordinate environmental observations across multiple sites.
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Alignment with Smart Docklands Objectives

SmartGreen Dock fully supports Smart Docklands’ mission to fast-track innovation addressing local and global challenges through collaboration, testing, learning, and education. The initiative transforms smart city infrastructure into a tool for empowerment—engaging citizens, informing decision-makers, and generating measurable environmental and social outcomes within six months. It represents a practical and scalable model for greener, smarter urban living that can evolve well beyond the Docklands.

Class Diagram



GitHub repo: [git@github.com:gavinmurphy09/Data_structures_Project.git](https://github.com/gavinmurphy09/Data_structures_Project.git)